

Gautham Bellamkonda

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RESEARCH INTERESTS

Complexity Theory: classical computational complexity; arithmetic and boolean circuits; boolean function complexity; communication complexity; fine-grained complexity.

Machine Learning Theory: computational learning theory; differential privacy; characterizing expressivity of modern architectures using complexity theory.

Quantum Computing: quantum speedups for classical problems; quantum query complexity; polynomial and adversary methods for finding upper and lower bounds on quantum query complexity.

EDUCATION

Indian Institute of Technology Hyderabad

B.Tech in Computer Science and Engineering, GPA: 8.95/10

Hyderabad, India

2020-2024

PROFESSIONAL EXPERIENCE AND INTERNSHIPS

KLA 

ML Research Intern, Algorithms Engineer

Chennai, India

Jan 2023 – Jul 2023, Jul 2024 – Present

StoryXpress 

ML Research Intern

Hyderabad, India

May 2022 – Jul 2022

RESEARCH PROJECTS

Logic Enhanced Concept Bottleneck Models

Advisor: Vineeth N Balasubramanian

IIT Hyderabad

Apr 2024 – Nov 2024

- Developed a framework integrating symbolic logic reasoning with concept bottleneck models to enhance interpretability and controllability in concept bottleneck classifiers.
- Conducted experiments on standard datasets, demonstrating improved performance and robustness against spurious correlations compared to traditional CBMs.

Lower and Upper Bounds in Quantum Query Complexity

Advisor: KartEEK Sreenivasaiah

IIT Hyderabad

Jan 2024 – May 2024

- Investigated quantum query lower bounds using the adversary method, deriving new insights based on Ambainis's techniques.
- Studied how combinatorial graph structures inform the design and efficiency of quantum algorithms for a few graph problems.

PUBLICATIONS

- Deepika Vemuri, **Gautham Bellamkonda**, Vineeth N Balasubramanian. **LogicCBMs: Using Logic to Enhance Concept-Based Learning**. In WACV 2026, Arizona, USA. (Accepted)

TEACHING EXPERIENCE

- **NPTEL Online Course:** Theory of Computation (Spring 2024)
- **IIT Hyderabad:** Discrete Mathematics (Fall 2021), Database Management Systems (Spring 2022), Computer Architecture (Fall 2022), Foundations of Machine Learning (Fall 2023)

SELECTED AWARDS AND HONORS

- Secured a Silver Medal in Math Bowl at the Inter-IIT Tech Meet 2023, representing IIT Hyderabad
- JEE Advanced: Ranked 419 in General Category out of 1,50,000+ candidates in 2020
- JEE Mains: Attained a rank of 243 in General Category out of 10,00,000+ candidates in 2020
- Qualified Indian National Mathematical Olympiad (INMO) in 2019 & 2020. Consequently attended IMOTCs.
- KVPY SA Fellow with AIR 53 out of 1,00,000+ candidates in 2019
- NTSE Scholar: Qualified Stages 1 & 2 of National Talent Search Examination in 2018

REFERENCES

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| <ul style="list-style-type: none">• Prof. Vineeth N Balasubramanian
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Indian Institute of Technology, Hyderabad
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